

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

MULTIDIMENSIONAL SCALING

SANJOY DASGUPTA, PROFESSOR

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE



Distance-Preserving Embeddings

- *Input:* An $n \times n$ matrix of pairwise distances

D_{ij} = desired distance between points i and j ,

as well as an integer k .

- *Desired output:* An embedding $z^{(1)}, \dots, z^{(n)} \in \mathbb{R}^k$ that realizes these distances as closely as possible.

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Most widely-used algorithm: **classical multidimensional scaling**.

- $D \in \mathbb{R}^{n \times n}$ matrix of desired *squared* interpoint distances.
- Schoenberg (1938): D can be realized in Euclidean space if and only if $B = -\frac{1}{2}HDH$ is positive semidefinite, where $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T$
- In fact, matrix B yields an approximate embedding even if it is not p.s.d.

Moving between Distances and Dot Products

Let $z^{(1)}, \dots, z^{(n)}$ be points in Euclidean space.

- **"Distance matrix":** Let $D_{ij} = \|z^{(i)} - z^{(j)}\|^2$ be the squared interpoint distances.
- **"Gram matrix":** Let $B_{ij} = z^{(i)} \cdot z^{(j)}$ be the dot products.

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- Can express squared distances using dot products.

$$\|x - x'\|^2 = \|x\|^2 + \|x'\|^2 - 2x \cdot x' = x \cdot x + x' \cdot x' - 2x \cdot x'.$$

Therefore, $D_{ij} = B_{ii} + B_{jj} - B_{ij} - B_{ji}$.

- Can express dot products in terms of distances, if points are centered at the origin.

$$B = -\frac{1}{2}HDH \text{ for } H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T$$

Question

Consider the two points $x_1 = (1,0)$ and $x_2 = (-1,0)$ in $\mathbb{R}^2$.

- 1 What is the matrix D of squared interpoint distances?
- 2 Write down $H = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^T$,
- 3 Compute $B = -\frac{1}{2}HDH$. Is this the correct Gram matrix?

Recovering an Embedding from Interpoint Distances

$D \in \mathbb{R}^{n \times n}$: matrix of desired *squared* interpoint distances.

Suppose these are realizable in Euclidean space: that is, there exist vectors $z^{(1)}, \dots, z^{(n)}$ such that $D_{ij} = \|z^{(i)} - z^{(j)}\|^2$.

We have seen how to obtain the Gram matrix, $B_{ij} = z^{(i)} \cdot z^{(j)}$.

- B is p.s.d. (why?). Thus its eigenvalues are nonnegative.
- Compute spectral decomposition:

$$B = U\Lambda U^T = YY^T,$$

where Λ is the diagonal matrix of eigenvalues and $Y = U\Lambda^{1/2}$.

- Denote the rows of Y by $y^{(1)}, \dots, y^{(n)}$. Then

$$y^{(i)} \cdot y^{(j)} = (YY^T)_{ij} = B_{ij} = z^{(i)} \cdot z^{(j)}.$$

If dot products are preserved, so are distances.

Result: embedding $y^{(1)}, \dots, y^{(n)}$ exactly replicates D . **What is the dimensionality?**

Classical Multidimensional Scaling

A slight generalization works even when the distances cannot necessarily be realized in Euclidean space.

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Given $n \times n$ matrix D of squared distances, target dimension k :

- 1 Compute $B = -\frac{1}{2}HDH$.
- 2 Compute the spectral decomposition $B = U\Lambda U^T$, where the eigenvalues in Λ are arranged in decreasing order.
- 3 Zero out any negative entries of Λ to get Λ_+ .
- 4 Set $Y = U\Lambda_+^{1/2}$.
- 5 Set Y_k to the first k columns of Y .
- 6 Let the embedding of the n points be given by the rows of Y_k .

Example

CITIES	ATLA	CHIC.	DENV.	HOUS.	L.A.	MIAMI	N.Y.	S.F.	SEAT.	WASH D.C.
ATLANTA		587	1212	701	1936	604	748	2139	2182	543
CHICAGO	587		920	940	1745	1188	713	1858	1737	597
DENVER	1212	920		879	831	1726	1631	949	1021	1494
HOUSTON	701	940	879		1374	968	1420	1645	1891	1220
LOS ANGELES	1936	1745	831	1374		2339	2451	347	959	2300
MIAMI	604	1188	1726	968	2339		1092	2594	2734	923
NEW YORK	748	713	1631	1420	2451	1092		2571	2408	205
SAN FRANCISCO	2139	1858	949	1645	347	2594	2571		678	2442
SEATTLE	2182	1737	1021	1891	959	2734	2408	678		2329
WASHINGTON D.C.	543	597	1494	1220	2300	923	205	2442	2329	

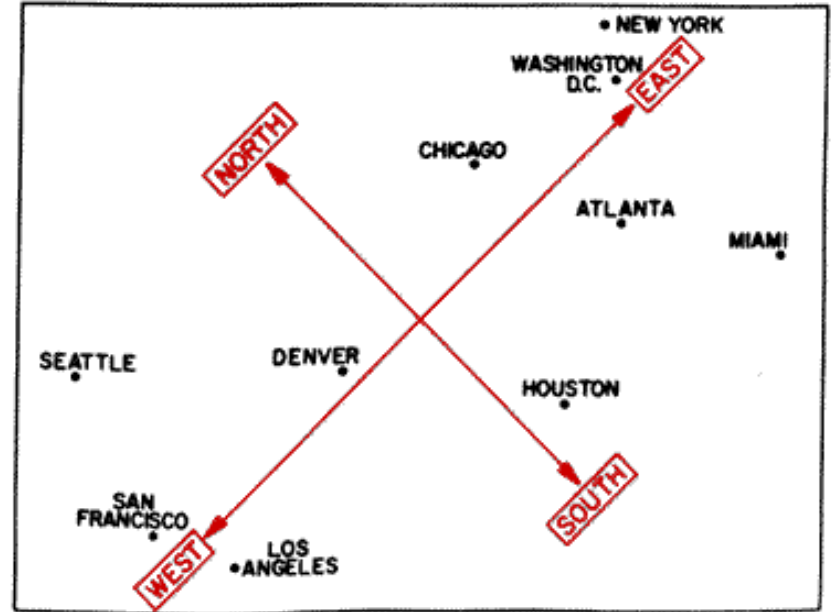


Figure: J.B. Kruskal and M. Wish (1978). Multidimensional scaling.

Alternative: Embedding by Gradient Descent

Given target distances $\{D_{ij}\}$, write down a cost function for the desired embedding $y_1, \dots, y_n \in \mathbb{R}^k$, e.g.,

$$L(y_1, \dots, y_n) = \sum_{i,j} |D_{ij}^2 - \|y_i - y_j\|^2|$$

and use gradient descent to optimize the $\{y_i\}$ directly.

Problem: no guarantees available.